

Homework #2 A Sample Solutions Set

Question 1. Perform the pairwise disjointness test for the following grammar rules.

$A \rightarrow aA \mid Bb \mid CBB$

$B \rightarrow aBb \mid b$

$C \rightarrow a \mid baC \mid c$

Answer:

For the grammar rule $A \rightarrow aA \mid Bb \mid CBB$ $\{a\} \cap \{a, b\} \cap \{a, b, c\} \neq \emptyset$ so, this rule can't pass the pairwise disjointness test.

For the grammar rule $B \rightarrow aBb \mid b$ $\{a\} \cap \{b\} = \emptyset$ so, this rule can pass the pairwise disjointness test.

For the grammar rule $C \rightarrow a \mid baC \mid c$ $\{a\} \cap \{b\} \cap \{c\} = \emptyset$ so, this rule can pass the pairwise disjointness test.

Question 2. Eliminate the direct left-recursion for the following grammar rules.

$A \rightarrow Aab \mid ABba \mid CBB \mid BC$

$B \rightarrow Bb \mid aBb \mid b$

$C \rightarrow Cc \mid baC \mid a$

Answer:

For the grammar rule $A \rightarrow Aab \mid ABba \mid CBB \mid BC$ the direct left-recursion is eliminated as follows:

$A \rightarrow CBBA' \mid BCA'$

$A' \rightarrow abA' \mid BbaA' \mid \varepsilon$

For the grammar rule $B \rightarrow Bb \mid aBb \mid b$ the direct left-recursion is eliminated as follows:

$B \rightarrow aBbB' \mid bB'$

$B' \rightarrow bB' \mid \varepsilon$

For the grammar rule $C \rightarrow Cc \mid baC \mid a$ the direct left-recursion is eliminated as follows:

$C \rightarrow baCC' \mid aC'$

$C' \rightarrow cC' \mid \varepsilon$

Question 3. Show a complete parse, including the parse stack contents, input string, and action for the string $(id + id) * (id * id) \$$, according to the shift-reduce algorithm using the grammar and parse table given in class.

Answer:

Stack	Input	Action
0	$(id + id) * (id * id) \$$	S4
0 (4	$id + id) * (id * id) \$$	S5
0 (4 id 5	$+ id) * (id * id) \$$	R6
0 (4 F 3	$+ id) * (id * id) \$$	R4
0 (4 T 2	$+ id) * (id * id) \$$	R2
0 (4 E 8	$+ id) * (id * id) \$$	S6
0 (4 E 8 + 6	$id) * (id * id) \$$	S5
0 (4 E 8 + 6 id 5	$) * (id * id) \$$	R6
0 (4 E 8 + 6 F 3	$) * (id * id) \$$	R4
0 (4 E 8 + 6 T 9	$) * (id * id) \$$	R1
0 (4 E 8	$) * (id * id) \$$	S11
0 (4 E 8) 11	$* (id * id) \$$	R5
0 F 3	$* (id * id) \$$	R4
0 T 2	$* (id * id) \$$	S7
0 T 2 * 7	$(id * id) \$$	S4
0 T 2 * 7 (4	$id * id) \$$	S5
0 T 2 * 7 (4 id 5	$* id) \$$	R6
0 T 2 * 7 (4 F 3	$* id) \$$	R4
0 T 2 * 7 (4 T 2	$* id) \$$	S7
0 T 2 * 7 (4 T 2 * 7	$id) \$$	S5
0 T 2 * 7 (4 T 2 * 7 id 5	$) \$$	R6
0 T 2 * 7 (4 T 2 * 7 F 10	$) \$$	R3
0 T 2 * 7 (4 T 2	$) \$$	R2
0 T 2 * 7 (4 E 8	$) \$$	S11
0 T 2 * 7 (4 E 8) 11	$\$$	R5
0 T 2 * 7 F 10	$\$$	R3
0 T 2	$\$$	R2
0 E 1	$\$$	accept

Question 4. Convert the following EBNF to BNF:

$A \rightarrow B(b \mid c)C$
 $B \rightarrow a\{Bb\}$
 $C \rightarrow c[d]$

Answer:

$A \rightarrow BbC \mid BcC$
 $B \rightarrow a \mid aD$
 $D \rightarrow Bb \mid BbD$
 $C \rightarrow c \mid cd$